

Quasi-amorphous crystal

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Brittle plastic yielding is a salient feature of well-annealed glassy materials. Here we show that the same behavior is characteristic of perfect crystals after they experience mechanically driven elastic instability leading to massive nucleation of dislocations. We argue that such ‘preparation’ effectively converts an atomic configuration from crystalline to quasi-amorphous. To understand the nature of the subsequent mechanical response, which is reminiscent of quasi-brittle yielding we study an athermal model 2D crystal subjected to quasistatic loading. We show that the intermittent pre- and post-yield dislocation avalanches exhibit power law statistics with matching exponents. The computed value of these exponents is indicative of marginal stability.

The complexity of plastic behavior of crystalline solids is revealed by pre- and post-yield intermittent fluctuations accompanied by scale-free organization of dislocations [1–10]. Such features are also typical of amorphous plasticity [11–20]. To build a conceptual link between the two phenomena, we study in this Letter a relatively transparent prototypical example of a 2D square crystal subjected to quasistatic loading in athermal conditions.

To prepare a ‘generic’ crystal we start with a pristine dislocation-free configuration and bring the associated perfect crystal to the point of mechanical instability which is resolved through massive dislocation nucleation. We argue that such ‘preparation’ effectively converts an atomic configuration from crystalline to quasi-amorphous [21–23]. The subsequent plastic flow begins with a hardening stage, which involves a superposition of spatially localized dislocation rearrangements. This ‘microplasticity’ regime is terminated abruptly with a quasi-brittle yield leading to the formation of multi-grain pattern involving global shear bands. The post-yield plastic flow is characterized by a stress plateau with superimposed fluctuations representing broadly distributed dislocation avalanches. Similar quasi-brittle yield has been observed in well-annealed glasses [24–32]. Despite the absence of dislocations in amorphous plasticity, behind both phenomena is an evolving elastic incompatibility [33–37] with underlying elementary mechanical events represented by shear eigenstrains [38–40].

To reveal the universal features of the criticality exhibited by crystal and amorphous plasticity, we perform a detailed study of dislocation dynamics in a prototypical athermal square crystal subjected to quasistatic loading by shear strain applied along one of the slip directions. Our main tool is a novel mesoscopic tensorial model (MTM) of crystal plasticity representing a conceptual trade-off between continuum and atomic descriptions [41–48]. Crucially, it allows one to capture the intermittent nature of dislocation avalanches while resolving topological transitions involved in nucleation and annihilation of dislocations without any specialized phenomenological assumptions [49]. In contrast to other computational approaches to micro-scale plasticity [50–75] the MTM allows one to deal with statistically meaningful number of dislocations while accounting adequately for large lattice rota-

tions and presenting geometrically exact description of lattice-invariant shears [76–79].

The main idea of the MTM approach is to associate with material points an effective energy landscape whose tensorial periodicity accounts for crystallographically-invariant deformations. The resulting Landau-type model is characterized by infinitely many equivalent energy wells [41–48, 80–90] and plastically deformed crystals emerge as a multi-phase mixtures with dislocations playing the role of effective domain boundaries. The small scale regularization is achieved by mesoscopic discretization. Despite operating with engineering concepts of stress and strain the resulting computational code correctly captures not only long-range elastic interactions of dislocations but also describes adequately their short-range interactions involved in the formation of complex dislocational entanglements. Some elements of the MTM approach can be found in [91–97], see also some parallel recent developments in [98–100].

Consider the continuum deformation field $\mathbf{y} = \mathbf{y}(\mathbf{x})$, where $\mathbf{y}$ and $\mathbf{x}$ are positions of material points in the current and reference configurations, respectively. Due to frame indifference constraint, the strain energy density φ can depend on the deformation gradient $\mathbf{F} = \nabla \mathbf{y}$, only through the metric tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ which then plays the role of a tensorial order parameter. The function $\varphi(\mathbf{C})$ should respect the symmetry of an underlying Bravais lattice extended beyond the conventional point group. For instance, to capture lattice invariant shears in 2D theory we should require that

$$\varphi(\mathbf{C}) = \varphi(\mathbf{m}^T \mathbf{C} \mathbf{m}), \quad (1)$$

with the unimodular integer valued matrix $\mathbf{m}$ belonging to $GL(2, \mathbb{Z})$, see [101] for details. In view of this symmetry, the surface $\det \mathbf{C} = 1$ in the 3D space (C_{11}, C_{22}, C_{12}) is tessellated into an infinite number of equivalent periodicity domains; if the energy density is known in one of such domains, it can be extended to the whole configurational space by $GL(2, \mathbb{Z})$ symmetry. To visualize this tessellation, it is convenient to project stereographically the infinite surface $\det \mathbf{C} = 1$ on a unit (Poincaré) disk, see Fig. 1 where we show in gray the fundamental (minimal periodicity) domain $\mathcal{D}$.

In our numerical experiments we used the polynomial function $\varphi(\mathbf{C})$ proposed in [89] which is the simplest to ensure

stress continuity, see [101] for details. In Fig. 2 we illustrate the corresponding multi-well energy landscape with equivalent energy wells representing replicas of the reference square lattice. One can see that such a landscape contains a network of low energy valleys depicting in our case simple shears parallel to crystallographic slip planes and therefore mimicking the conventional plastic 'mechanisms' operating in square crystals.

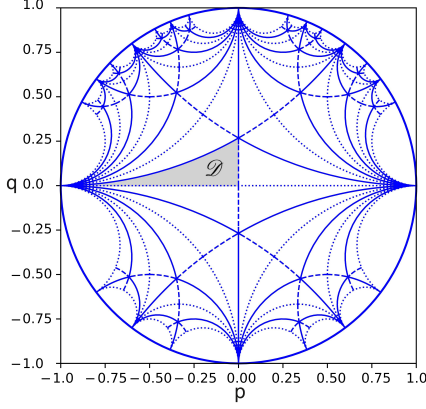


FIG. 1. $GL(2, \mathbb{Z})$ -induced tessellation of the configuration space of metric tensors C_{11}, C_{22}, C_{12} with $\det \mathbf{C} = 1$ stereographically projected on the Poincaré disk $p^2 + q^2 < 1$ where $p = t(C_{11} - C_{22})/2$, $q = tC_{12}$, and $t = 2/(2 + C_{11} + C_{22})$. The highlighted domain $\mathcal{D}$ is the fundamental (minimal periodicity) domain.

The presence of the energy barriers separating individual energy wells makes the function $\varphi(\mathbf{C})$ highly nonconvex, see Fig. 2. Therefore, the corresponding scale-free Landau model is degenerate and needs to be regularized through the introduction of a cut-off length scale. This is achieved in the MTM by projecting the continuum problem on a uniform mesoscale grid and associating the periodic elastic energy with discrete elements whose linear size h becomes a physical parameter. In this way the infinite dimensional space of continuum deformation fields is reduced to a finite dimensional set of compatible, piece-wise affine mappings, see [101] for details.

The driven system of this type will experience a rich repertoire of instabilities. Specifically, an incremental elastic energy minimization of a system subjected to quasistatic loading will generically lead to repeated jump discontinuities representing branch switching events. The energy losses associated with such jumps will represent plastic dissipation; in the continuum limit the jumps will merge giving rise to rate-independent plasticity [102].

In our numerical experiments a square crystal was represented by $N \times N$ nodes, with $N = 400$. The nodes were arranged into triangular finite elements. We loaded the corresponding athermal system quasi-statically by applying the affine displacement field

$$\mathbf{u}(\alpha, \mathbf{x}) = (\bar{\mathbf{F}}(\alpha) - \mathbb{1})\mathbf{x}, \quad (2)$$

with

$$\bar{\mathbf{F}}(\alpha) = \mathbf{I} + \alpha \mathbf{e}_1 \otimes \mathbf{e}_2,$$

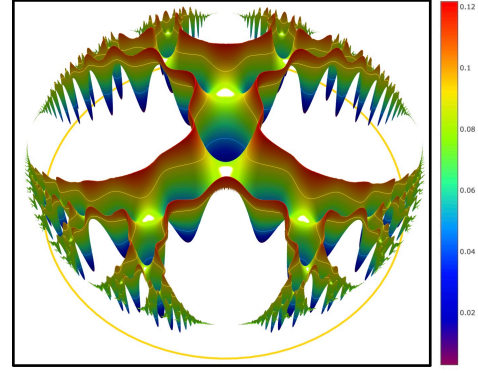


FIG. 2. Energy landscape in the space of metric tensors (Poincaré disk shown in Fig. 1). Color indicates the energy level with only the lowest energy levels shown. The deepest energy wells correspond to equivalent representations of the same square lattice.

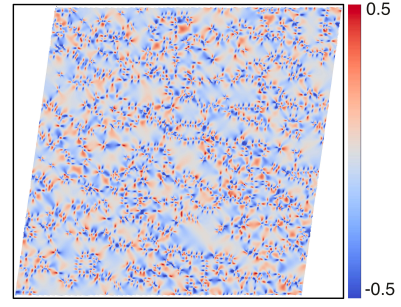


FIG. 3. Dislocated configuration emerging after a homogeneous lattice is mechanically driven to the threshold of elastic instability. Colors indicate the level of the shear component of the Cauchy stress tensor σ_{xy} . The highest stress levels of both signs correspond to the location of dislocation cores.

where $\mathbf{e}_i$, $i = 1, 2$, is the orthonormal basis of the reference square lattice. In this way we imposed simple shear along one of the principal slip directions with shear amplitude α serving as the loading parameter. By changing this parameter in strain increments of order 10^{-6} , we advanced the loading. After each such increment the displacement field was updated by the energy minimization algorithm. Details of the discretization, of the incremental energy minimization algorithm and of the implementation of the loading protocol under periodic boundary conditions can be found in [101].

To obtain a 'generic' sample, we first loaded a pristine (defect-free) crystal till the affine configuration became elastically unstable. The corresponding critical value of the loading parameter $\alpha = \alpha_c \approx 0.138$ is very close to the threshold of positive definiteness of the acoustic tensor [103–105], see the explicit formulas in [101]. The breakdown of an elastic state takes the form of a major system-size inelastic event. It leads to multiple nucleation of dislocations which apparently transforms a perfect crystal into a quasi-amorphous system, see Fig. 3. We observe that the unfolding of the elastic instability is arrested at a state which one can expect to be only

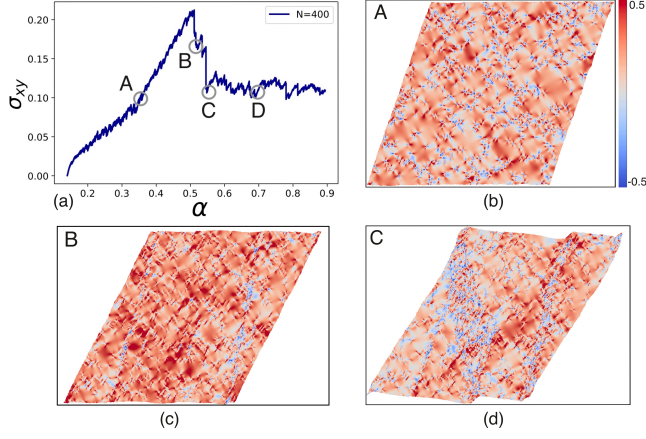


FIG. 4. (a) Stress-strain response of a quasi-amorphous crystal shown in Fig. 3. (b) Dislocated configuration in a typical pre-yield state characterizing the stage of 'microplasticity'. (c,d) Dislocated configuration characterizing different stages of unfolding of the quasi-brittle system size event. Colors indicate the level of the shear component of the Cauchy stress tensor σ_{xy} .

marginally stable.

We continued to load the obtained 'generic' sample using the same athermal quasistatic protocol and the ensuing stress-strain response is shown in Fig. 4(a). Its salient feature is the presence of irregularly placed elastic branches interrupted by intermittent stress drops; the latter represent plastic avalanches involving partial unlocking of dislocation structures. Among these avalanche-type events, one major stress drop stands out and can be interpreted as a plastic yield separating two different regimes. While the first (pre-yield) regime is characterized by quasi-elastic hardening, the second (post-yield) regime is represented by a stress plateau which suggests quasi-stationary plastic flow, see Fig. 4(a).

Our Fig. 4(b) illustrates the stress field in a typical pre-yield state A. The restructuring of the dislocation pattern is relatively minor with dislocations moving only between the pre-existing locking sites, cf. Fig. 3. Such response is known to be a characteristic pre-yield feature for both amorphous and (defective) crystalline solids [4, 106–110]. This 'microplasticity' regime ends with the system-size event reminiscent of quasi-brittle plastic yield encountered in well-annealed glasses and also in some sub-micron crystals [111–121]. As we see from Fig. 4(a) it takes the form of a two-stage transition $A \rightarrow B \rightarrow C$; the snapshots of the spatial stress configurations in the states B and C are shown in Fig. 4(c,d). The associated global restructuring reveals collective dislocation activity manifested through the emergence of system-spanning shear bands.

Additional details can be seen in Fig. 5 where we illustrate the strain-energy density field in the state C. We see the development of low-energy rotated patches of the original lattice forming intricate polycrystalline grain texture with misorientation of neighboring grains controlled by the overall compatibility of the deformation field as explained in [47]; the elastic energy is localized around dislocation-rich grain

boundaries. In the inset in Fig. 5 we made dislocations of different sign clearly visible: they were identified via Delaunay triangulation and nodes with five/seven neighbors are shown in blue/red. One can see the emergence of energy minimizing structure of grain boundaries [122–124] with ubiquitous presence of $\Sigma 5$ grain boundaries, see [47] for the explanation.

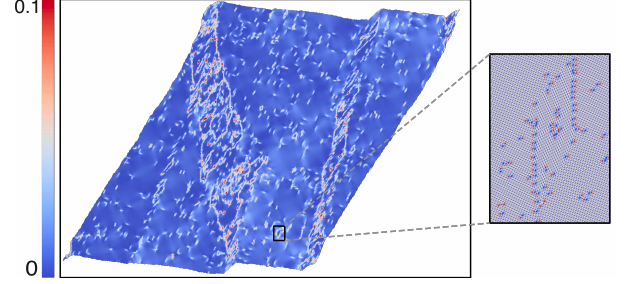


FIG. 5. The micro-configuration configuration corresponding to the state C marked in Fig. 4. Colors indicate the magnitude of strain-energy density. The inset shows a single grain and details the dislocation structure of the grain boundaries.

The post-yield regime, characterized by relative stabilization of the average stress level, can be interpreted as a quasi-stationary non-equilibrium steady state (NESS). A representative snapshot of the micro-configuration of the crystal in the state D is shown in Fig. 6 where we see further development of the polycrystalline grain texture. During intermittent avalanches characterizing this regime larger grains occasionally merge while small grains continue to emerge. In this way the self-organization of dislocations continues to generate additional scales which contribute to the formation of the global hierarchical structure. Through the transitions from isolated dislocation motion to collective dislocation behavior the system acquires access to a much broader repertoire of relaxation mechanisms. A detailed analysis of the avalanche structure shows that while in the pre-yield regime plastification takes place primarily in the form of isolated, linear arrangements of transformed elements, a typical post-yield avalanche reveals extended plastified regions with complex branching.

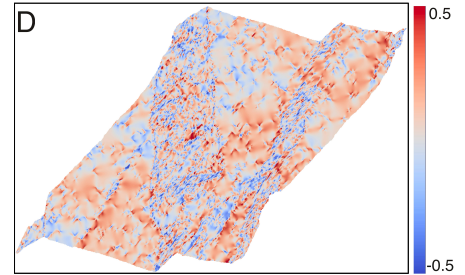


FIG. 6. Mature shear bands in the regime of stationary post-yield plastic flow represented by the state D marked in Fig. 4. Colors indicate the level of the shear component of the Cauchy stress tensor σ_{xy} .

An important window into the micro-mechanics of solids

undergoing plastic flow is provided by the study of the statistical structure of plastic fluctuations [1, 4, 22, 125–128]. Thus, it is known that in amorphous solids the probability distribution of the energy drops associated with individual avalanches $\Delta W = W_{\text{before}} - W_{\text{after}}$, where W is the total elastic energy of the crystal defined in [101], is represented by power laws of the form

$$P(x) \sim x^{-\epsilon} e^{-\lambda x} \quad (3)$$

with the same exponent ϵ but different values of the cut-off parameter λ for pre-yield and post-yield regimes [39, 129–131]. Our results, illustrated in Fig. 7, show that for ‘generic’ crystals prepared using our protocol, the statistics of avalanches follows the same pattern. In particular, similar to the case of well-annealed glasses [4, 21, 49, 62, 128, 132], the scaling behavior remains basically unchanged across the quasi-brittle yielding transition. The obtained power law range extends over up to six decades with pre-yield and post-yield energy distributions sharing the same exponent $\epsilon = 1.0$. The invariance of the exponent ϵ across the yielding transition suggests that while the spatial organization of plastic events changes drastically from local and non-cooperative to global and collective, the statistical distribution of events remains insensitive to these details responding only to the global organization of the underlying energy landscape [13, 133].

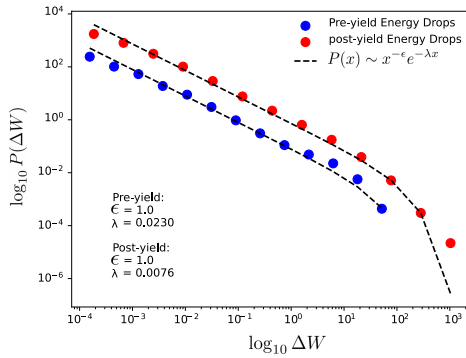


FIG. 7. Probability distribution of energy drops during avalanches in pre- and post-yield regimes.

The value of the power-law exponent $\epsilon = 1.0$ is a signature of an archetypically ‘wild’ crystal plasticity in the sense of [125]. The same value was obtained in discrete dislocation dynamics studies of disorder-free 2D samples containing a fixed number of preexisting dislocations [62, 134] and was also recorded in numerical experiments on pristine crystals using a scalar version of the MTM [48]. The exponent $\epsilon = 1.0$ has been previously interpreted in crystal plasticity framework as representing either dislocation jamming or self-induced glassiness [21, 135, 136].

It is rather striking that exactly the same value of the exponent has been also recorded in the studies of plastic flows in structural glasses [13, 137, 138]. A theoretical understanding of the relation between the emergence of the power law exponent $\epsilon = 1.0$ and the structure of the underlying energy landscapes has been developed in the theory of spin glasses.

Thus, it was shown that the hierarchical (ultrametric) organization of energy wells in the phase space results in an intermittent, scale-free response to quasistatic deformation [139–142]. Moreover, it was rigorously proved that in the mean-field limit the associated distribution of energy avalanches must be of a power law with exponent $\epsilon = 1.0$. In the spin glass context it has been also understood that the reason behind this particular value of the exponent is the marginal stability of the system [139, 143, 144].

Since glasses can be viewed as liquids at a quenching threshold, their marginal stability is similar to marginality exhibited by granular matter at a jamming point. If such systems are mechanically driven, the proximity to unstable modes leads to the mixing of statics (stability) and dynamics (instability) making the mechanical response inherently intermittent [145, 146]. Similarly, since our ‘quasi-amorphous crystal’ emerged from arrested dynamics, one can argue that the associated self-generated disorder brought the system from a solid to an effectively glassy state. Furthermore, our results suggest that the implied marginality is not affected by the quasi-brittle yield, which apparently does not compromise the global organization of the energy landscape while, of course, affecting the reachability of its different subdomains [147–149].

Note next that the pre-yield value of the cutoff parameter $\lambda = 0.0230$ is drastically different from its post-yield value $\lambda = 0.0076$. The approximately three-fold decrease of the value of λ indicates that large energy drops become substantially more probable in the post-yield regime [18, 150]. The emergent collective behavior is reflected in a much broader extent of avalanche activity which reaches the size of the system suggesting the divergence of the characteristic length $l \sim 1/\lambda$. The physical nature of the implied criticality is presently actively debated [48, 151–155].

Finally we mention that the observed similarity between the structure of intermittent plastic fluctuations in amorphous and crystalline solids is supported by the fact that elasto-plastic models, typically used to model amorphous plasticity [24, 138, 156, 157], are very similar to the MTM which operates within basically the same finite element setting. The difference is that the phenomenological yield thresholds of elasto-plastic models are replaced in the MTM by elastic instabilities originating from the non-convexity of the energy landscape. Accordingly, the fixed linear elastic propagators are replaced by the solution ‘on the fly’ of the corresponding nonlinear elasticity problems. It is important that the crucial nonlinearity in the MTM is of universal geometrical nature as it originates from the exact description of finite elastic deformations.

To conclude, we showed that mechanically driven perfect crystals can exhibit quasi-brittle plastic yielding which is remarkably similar to the behavior of well-annealed glassy materials. The implied parallel features of the mechanical response emerge after pristine crystals acquire self-generated disorder resulting from massive dislocation nucleation during the breakdown of an affine elastic configuration. The ensuing quasi-amorphous crystals exhibit pre- and post-yield avalanches with power law statistics whose matching exponents are indicative of the marginality of the associated me-

chanical system. Adaptation of the same model to realistic 3D crystals will allow one to distinguish between edge and screw dislocations opening the way to the adequate account of such physically important effects as dislocation climb, cross slip and forest hardening.

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